ESC-201 Mechanics of Materials Notes

The main idea of this course is to understand how mechanisms and designs perform under various loads and conditions. We will combine force loading from statics and dynamics with material properties to quantify the performance of a design and see how it fails.

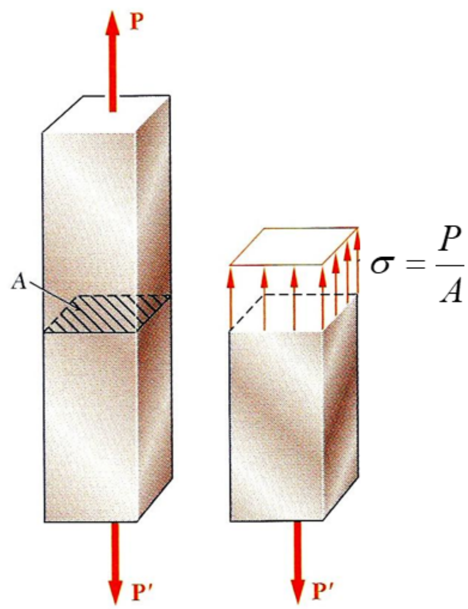
Basic Concepts

Normal Stress

When load is applied normal to a cross-section such as the one below, we can find the normal stress applied by dividing the force by the area of the cross-section. This represents the *average* normal stress, as the actual stress distribution is usually no uniform across the cross-section. It can be thought similarly to pressure, which both represents a load over an area.

$$\sigma_{\text{avg}} = \frac{P}{A}$$

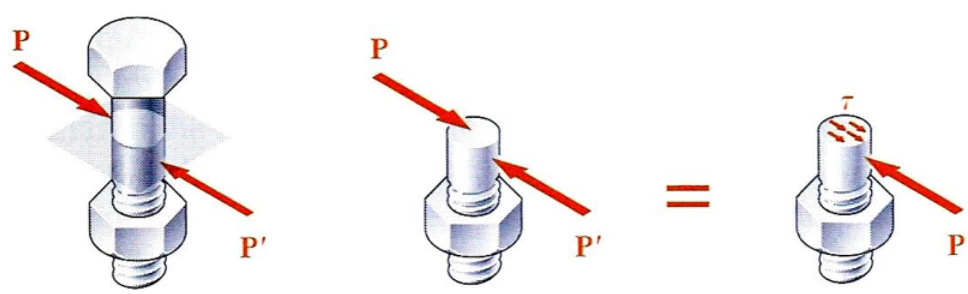
Sign convention: A positive stress corresponds to **tensile** loading while a negative stress corresponds to **compressive** loading.



Shear Stress

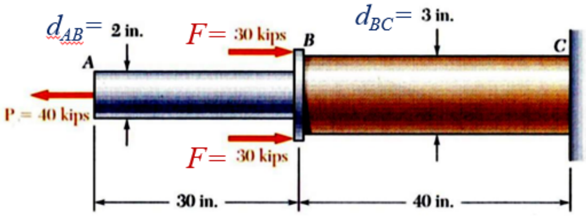
When load is applied parallel or along a cross-section, we can find the shear stress by dividing the force by the area of the cross-section. Similar to normal stress, this represents the *average* shear stress, as the actual stress distribution is usually not uniform across the cross-section.

$$\tau_{\text{avg}} = \frac{P}{A}$$

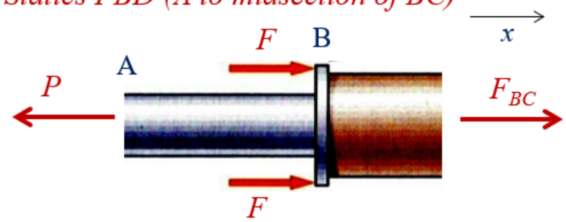


Sign convention: It follows the sign of the applied force. (ie. If the force is in the positive direction, the shear stress is positive.)

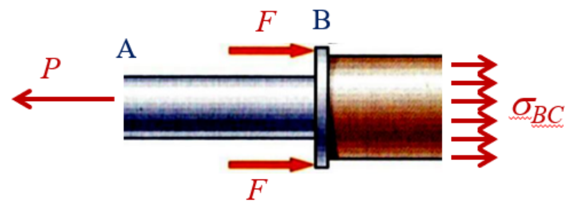
Note: It is sometimes useful to think of stress as replacement for forces. For example, we can draw two different FBDs, one showing the forces and one showing the stresses.



Statics FBD (A to midsection of BC)

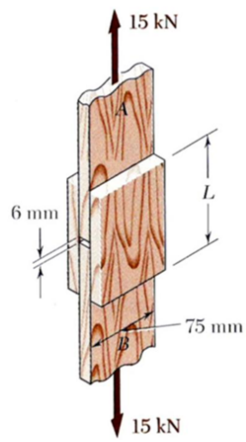


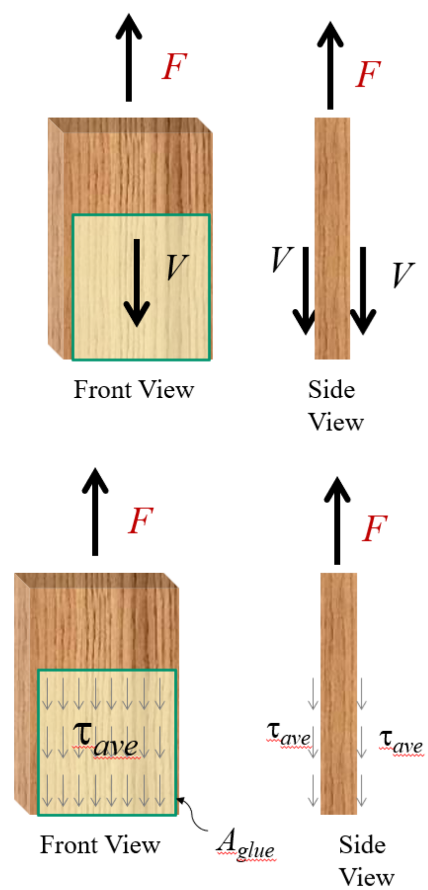
MechMat FBD (replace F_{BC} with σ_{BC})



This way, it is clear to see what force corresponds to what stress and what area it acts on.

This is representative of our problem-solving approach. Starting from static force balances to solve for the forces, then using those forces to find the stresses, and finally using the stresses to find the strains and deformations.





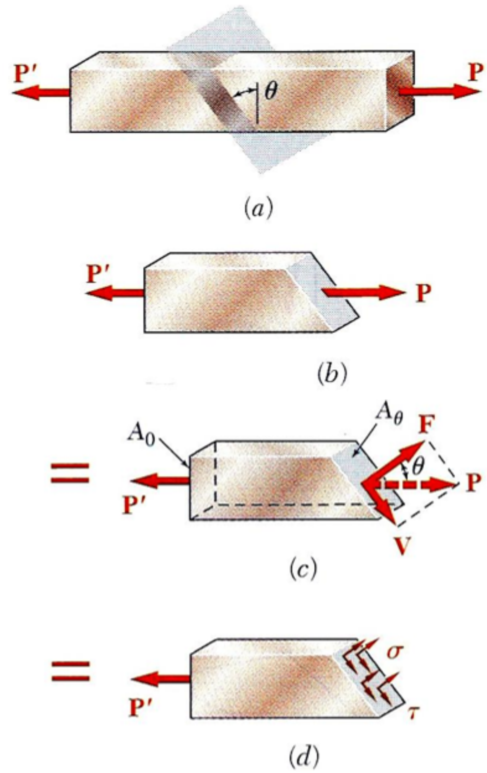
General Stress

When either the load is applied at an angle or the cross-section is not perpendicular to the load, we can always decompose the load into a normal and shear component. Then we can define the normal and shear stresses as before.

$$\sigma = \frac{F}{A_\theta} = \frac{Pcos\theta}{A_0/cos\theta} = \frac{Pcos^2\theta}{A_0}$$

$$\tau = \frac{V}{A_\theta} = \frac{Psin\theta}{A_0/cos\theta} = \frac{Psin\theta cos\theta}{A_0}$$

Where A_0 is the area of the cross-section perpendicular to the load/axis and A_θ is the area of the cross-section at an angle θ to the load/axis.



Pinned Joints

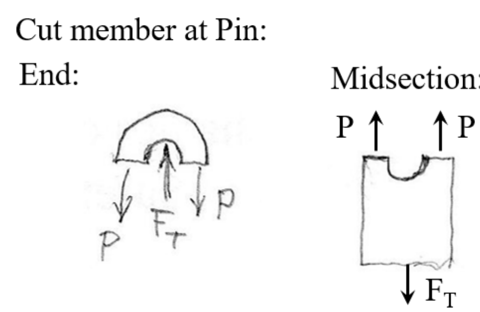
When considering a pinned joint and how it fails, there are several ways. Mainly, it depends on whether the load is tensile or compressive. For both cases, the average normal stress at the midsection matters, but it's not usually where failure *first* occurs, or in other words, it's not a critical region (it is usually where stress is max). The following are the critical regions unique to the type of loading:

- **Member in Tension**

  Under tension, failure can occur by the member splitting at the shoulders around the pin where the cross-sectional area is smallest.

  $$\sigma_{\text{avg}_{\max}} = \frac{F_T}{A_{\min}} \quad \text{(max average normal stress)}$$

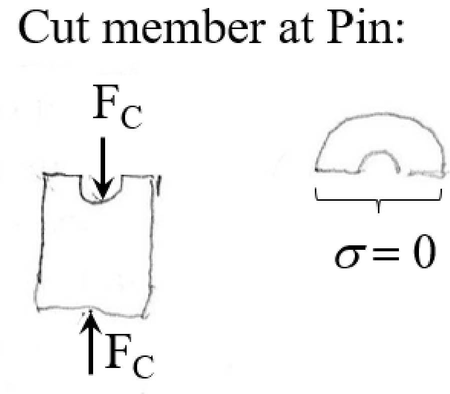
  Where $A_{\min}$ is the minimum cross-sectional area of the member, which is usually defined by $(w-d)t$ where w is the width of the member, d is the diameter of the pin, and t is the thickness of the member.



- **Member in Compression**

  Under compression, the pin squishes into the member and effectively becomes a solid piece. Thus, the risk isn't at the shoulders but rather at the midsection where it could buckle. Note that there is ZERO stress at the upper region around the hole since there is no load there.

  $$\sigma_{\text{avg}} = \frac{F_C}{A_{\text{cross-section}}} \quad \text{(the usual average normal stress)}$$



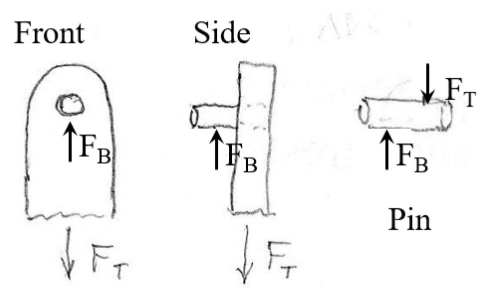
For both cases, we need to also consider the shear stress on the pin and also the bearing stress on the member.

Shear Stress at Pin

The first place failure can occur is that the pin shears. In this case, the pin is in *single shear*. This is the same for both tension and compression.

$$\tau_{\text{avg}} = \frac{F_T}{A_{\text{pin}}}$$

Where A_{pin} is the cross-sectional area of the pin.



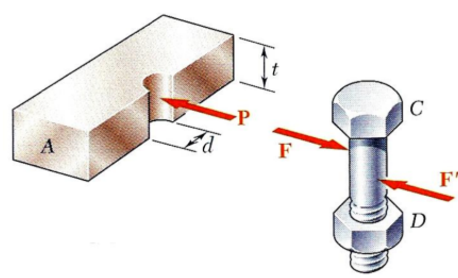
Bearing Stress

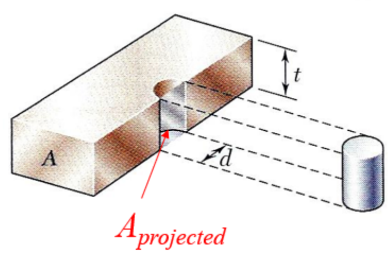
Another place failure can occur is that the hole the pin rests upon physically deforms and causes failure. This is called bearing stress, and it is always a compressive stress (the pin pushes into the hole).

It is also an average stress, but it uses the projected area of the bolt onto the hole:

$$\sigma_{\text{avg}} = \frac{P}{A_{\text{projected}}} = \frac{P}{td}$$

Sign convention: Like mentioned before, this is always a compressive stress, so we report it as a positive value.





Factor of Safety and Stresses

Factor of Safety

We use the factory of safety to compensate for using uniform stress assumptions and other variability. It is defined as

$$\text{FS} = \frac{\text{Ultimate Load}}{\text{Allowable Load}}$$

Where the ultimate load is the max load at failure and the allowable load is the max design load. If failure is due to stress exceeding the ultimate strength, then

$$\text{FS} = \frac{\sigma_{\text{ult}}}{\sigma_{\text{allowable}}}$$

Where σ_{ult} is the ultimate strength of the material and $\sigma_{\text{allowable}}$ is the allowable stress.

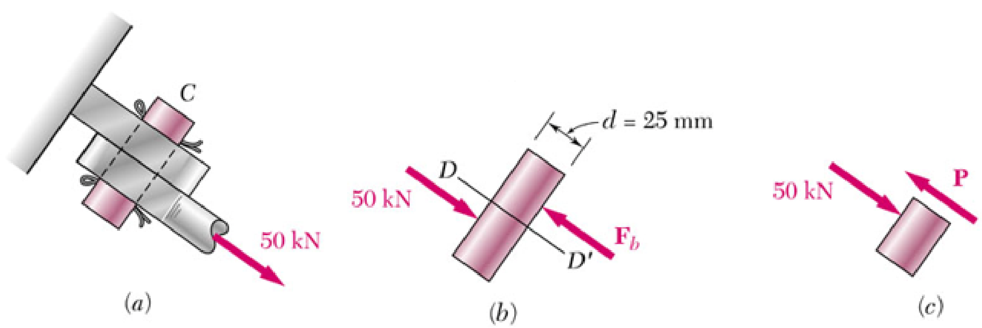
Shear in Pin or Bolt

We've already mentioned single shear as a mode of failure for pinned joints, but here it is again:

Single Shear

Single shear means that there is one member per side of the pin, such that the shear load is *equal* to the member/applied load.

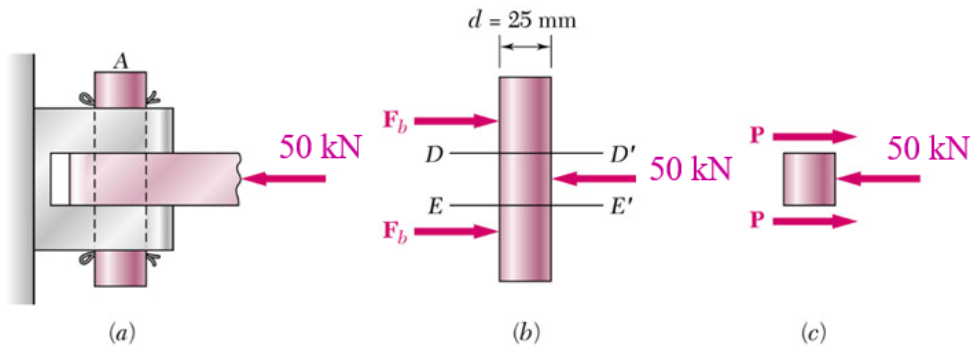
$$\tau_{\text{avg}} = \frac{\text{load}}{\text{pin XC area}}$$



Double Shear

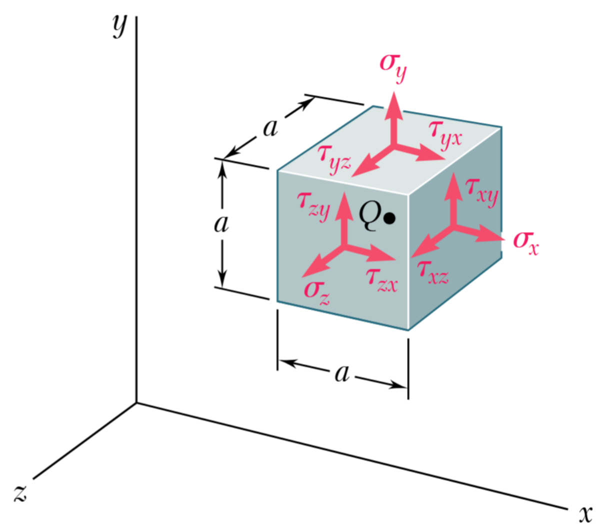
Double shear means shear is carried on both sides of incoming member (clevis joint) such that the load is split between two sides of clevis.

$$\tau_{\text{avg}} = \frac{\text{load}/2}{\text{pin XC area}}$$



3D State of Stress

Consider an infinitesimal cube of material that is internal to a member under load with side lengths a much smaller than the member. All the normal and shear stresses on the faces are *internal* stresses exerted by the surrounding material and not the applied load.



The cube is in equilibrium, so the forces on each face must balance out. Thus, we only need to define the stresses on the positive faces, and the stresses on the negative faces will be of equal magnitude but opposite direction.

There are 3 normal stresses σ_i

- i is the direction of the outward normal vector of the face (face direction)
- i is also the direction of the force (by definition of normal stress)

There are 6 shear stresses τ_{ij}

- i is the direction of the outward normal vector of the face (face direction)
- j is the direction of the force (perpendicular to the face direction by definition of shear stress)

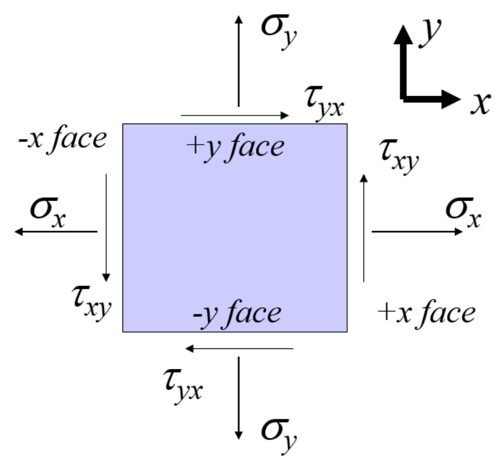
Note: Define a coordinate system and be consistent with the directions of the normal and shear stresses (point along the positive direction of the coordinate system).

Stress on the Negative Faces

The cube of material is in equilibrium, so the forces on each face must balance out. Since each face has the same area, the corresponding stresses must also balance.

$$\sum F_x = 0 \quad \Longrightarrow \quad \sigma_x - \sigma_x + \tau_{xy} - \tau_{xy} = 0$$

$$\sum F_y = 0 \quad \Longrightarrow \quad \sigma_y - \sigma_y + \tau_{xy} - \tau_{xy} = 0$$



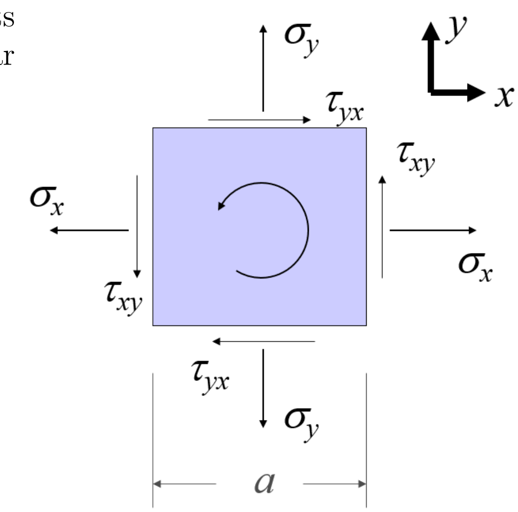
Shear Stress Symmetry

Building on the equilibrium of the cube, the moments must also balance out. Thus, we can define a shear stress symmetry:

$$\sum M_z = 0 \quad \Longrightarrow \quad 2\tau_{xy}\left(\frac{a}{2}\right)a^2 - 2\tau_{yx}\left(\frac{a}{2}\right)a^2 = 0$$
$$\Longrightarrow \quad \tau_{xy} = \tau_{yx}$$

Therefore,

$$\boxed{\begin{aligned} \tau_{yx} &= \tau_{xy} \\ \tau_{zx} &= \tau_{xz} \\ \tau_{zy} &= \tau_{yz} \end{aligned}}$$



Note: Shear stress symmetry holds even with acceleration and angular acceleration (non-equilibrium) since the moments from the inertial forces still balance out (as length $a \to 0$).

$$\sum M_z = I_{zz}\alpha$$
$$\sum M_z = 2\tau_{xy}\left(\frac{a}{2}\right)a^2 - 2\tau_{yx}\left(\frac{a}{2}\right)a^2$$
$$I_{zz} = \frac{1}{6}ma^2 = \frac{1}{6}\rho a^5$$
$$\Rightarrow a^3(\tau_{xy} - \tau_{yx}) = \rho a^5 \alpha$$

Limit as $a \to 0$

$$\tau_{yx} - \tau_{xy} = \rho a^2 \alpha \to 0$$
$$\tau_{yx} = \tau_{xy}$$

Axial Loading and Deformation

This is going to be a long section, we finally introduce strain and deformation, allowing us to find things such as thermal stress, and solve for statically indeterminate problems.

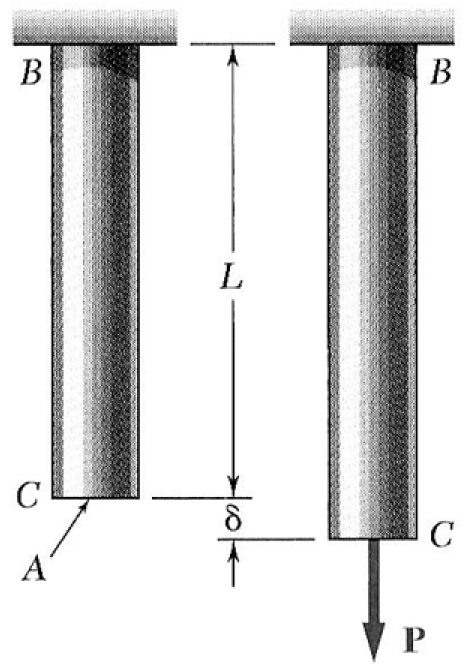
Axial Strain

If deflection δ is defined as the change in length of a member (usually by some load), then we can define the axial strain as the amount of deflection per unit length. It is a dimensionless quantity.

$$\boxed{\varepsilon = \frac{\delta}{L} \quad \text{(uniform strain)}}$$

For a general member, we can define the limit

$$\varepsilon = \lim_{\delta \to 0} \frac{\Delta\delta}{\Delta L} = \frac{d\delta}{dL} \quad \text{(local strain)}$$



Axial Deflection

In the elastic region below the yield stress, we can use Hooke's Law to relate the axial stress to the axial strain using the **Young's Modulus** or **modulus of elasticity** E.

$$\boxed{\sigma = E\varepsilon}$$

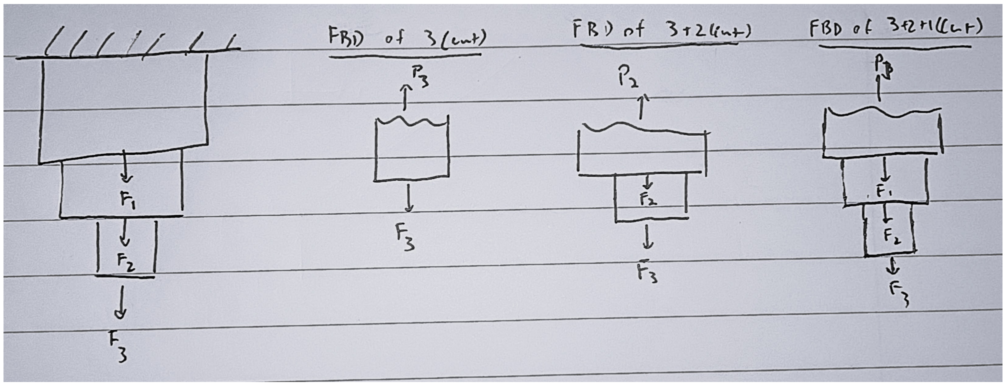
Thus, we can define the **axial deflection** as

$$\boxed{\delta = \frac{PL}{EA}}$$

and **stiffness** of the member as

$$\boxed{k = \frac{P}{\delta} = \frac{EA}{L}}$$

Note that P is an *internal* reaction force, not the applied load, so we must first solve for the internal forces using static force balances.



Thus, we can generalize for non-uniform members by integrating over the length of the member:

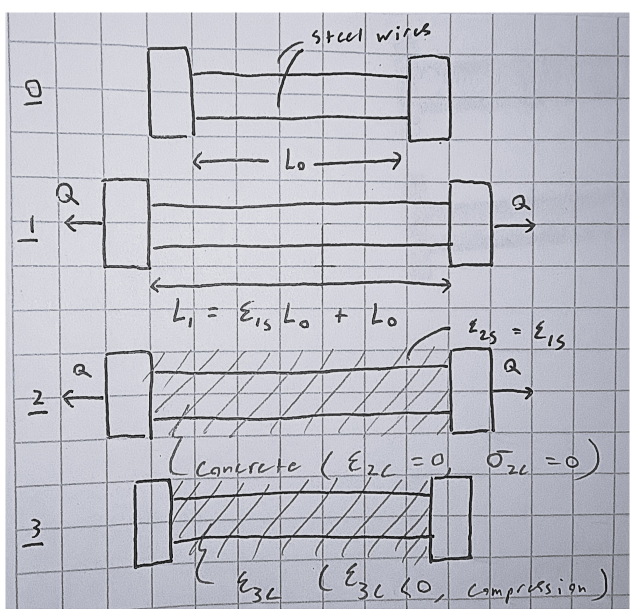
$$\delta = \sum \frac{P_i L_i}{E_i A_i} = \int_0^L \frac{P}{EA} dx$$

Statically Indeterminate Problems

The nice thing about having a deflection formula is that we can now apply constraints on the deflection to solve for unknown forces. A useful analogy here is to think of the members as springs. In some cases, the members are in series where the total length is constant ($\delta_1 + \delta_2 = 0$), and in some cases, the members are in parallel where the deflection is constant ($\delta_1 = \delta_2$). We can use these constraints to solve for unknown forces and then find the stresses and strains as usual.

Prestressed Concrete

Since concrete is stronger in compression than in tension, it is common to *preload* the concrete with a compressive force (from embedded steel wires). It works by the following process:



Initially (state 0), there is no concrete and the steel wires are at length L_0.

1. Pre-tensioning: The steel wires are stretched to a length L_1 by a load Q and then fixed in place. This creates a strain
$$\varepsilon_{1s} = \frac{Q}{A_s E_s}$$

2. Casting: While the wires are held under tension, the concrete is poured around them and allowed to set (harden). At this moment, the concrete has zero stress and zero strain, but it is now bonded to the stretched wires.

3. Release: The external load Q is removed. The steel wires want to shrink back to their original length, but the hardened concrete is in the way. Thus, as the steel contracts slightly, it pushes against the concrete, putting the concrete into compression.

Thus, the strain of the steel and concrete in state 3 must satisfy this constraint equation, with the concrete strain negative due to compression and the steel strain positive due to tension.

$$\boxed{\varepsilon_{3s} = \varepsilon_{2s} + \varepsilon_{3c}}$$

To determine the strains, we use static equilibrium and express the loads in terms of Hooke's law.

$$P_s + P_c = 0$$

$$P_s = E_s A_s \varepsilon_{3s} = E_s A_s (\varepsilon_{2s} + \varepsilon_{3c}), \quad P_c = E_c A_c \varepsilon_{3c}$$

Thermal Stress

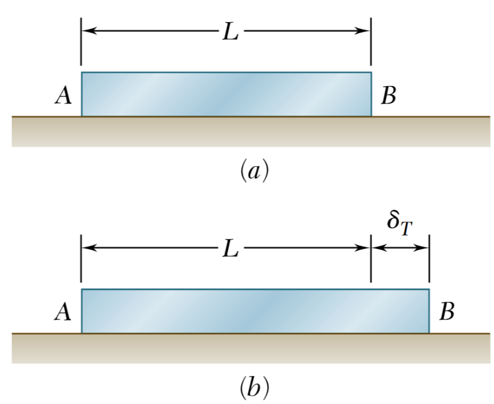
Thermal Expansion

When an unconstrained member is heated, it expands by a length δ_T that is given by the coefficient of thermal expansion α and the change in temperature ΔT.

$$\boxed{\delta_T = \alpha(\Delta T)L}$$

The thermal strain is then

$$\boxed{\varepsilon_T = \frac{\delta_T}{L} = \alpha \Delta T}$$



Thermal Stress

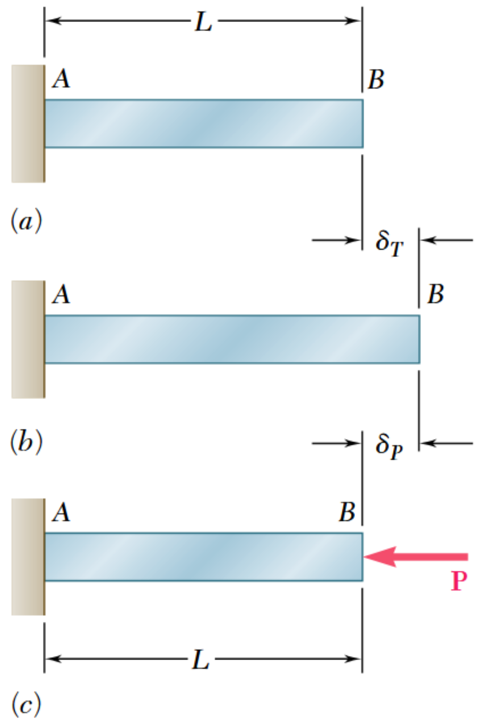
To determine the thermal stress in a *constrained member*, we use a superposition approach similar to the mechanics of prestressed concrete. The process is modeled in two stages:

1. Free Expansion: The member is imagined to undergo unconstrained thermal expansion

$$\delta_T = \alpha(\Delta T)L$$

2. Constraint Application: A compressive load P is then applied to provide a mechanical displacement δ_P that restores the member to its original length.

$$\delta_P = -\frac{PL}{EA}$$



Since the net displacement in a fixed, constrained structure is zero, we have the constraint

$$\boxed{\delta = \delta_T + \delta_P = 0}$$

Applying the constraint to solve for the induced force:

$$\alpha(\Delta T)L = \frac{PL}{EA} \implies P = \alpha(\Delta T)EA$$

The resulting stress from this restorative load corresponds to the thermal stress in the member:

$$\boxed{\sigma_T = \frac{-P}{A} = -\alpha(\Delta T)E}$$

Sign convention: For a positive change in temperature (heating), the thermal stress is compressive (negative) while for a negative change in temperature (cooling), the thermal stress is tensile (positive).

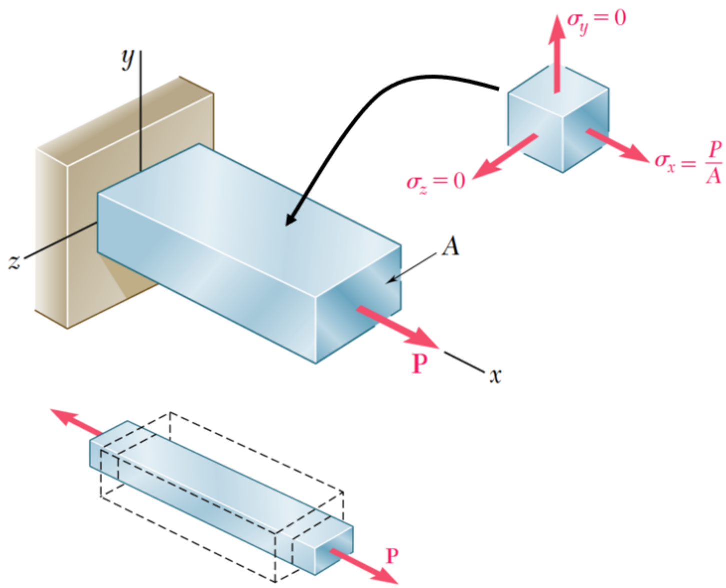
Poisson's Ratio

When a member is stretched in one direction, it causes transverse shrinkage in most materials. This is quantified by Poisson's ratio ν. In the case of *uniaxial* stress:

$$\sigma_x = \frac{P}{A}, \quad \sigma_y = \sigma_z = 0$$

$$\varepsilon_x = \frac{P}{EA}$$

$$\boxed{\varepsilon_y = \varepsilon_z = -\nu\varepsilon_x}$$



Saint-Venant's Principle

So far, we have considered only *uniform* stress and strain or calculated the *average* values. In reality, point loads and geometric discontinuities (holes, notches, etc.) produce non-uniform stress and strain distributions.

Saint-Venant's principle identifies the regions where the average stress and strain assumptions hold and states that the influence of these *stress concentrators* can be expressed as the average stress or strain multiplied by a certain factor.

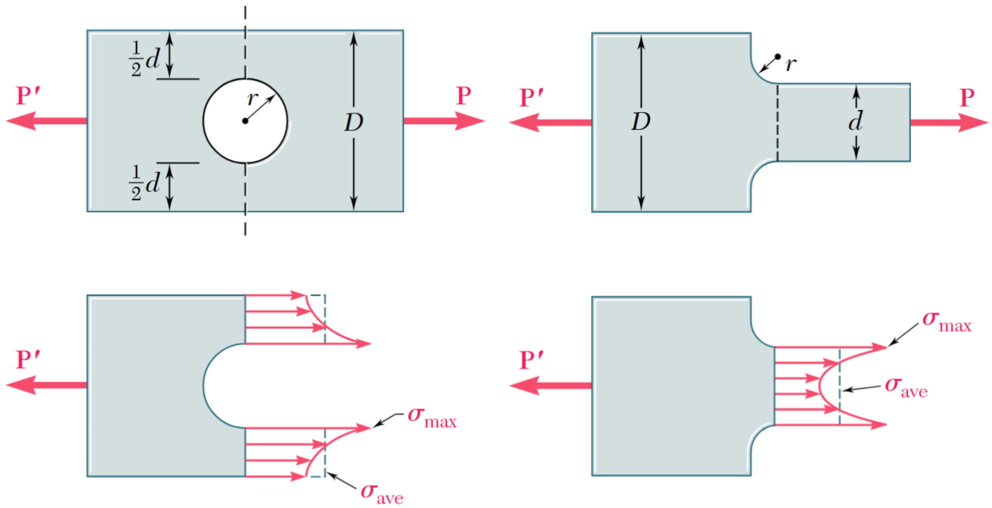
Specifically, in the region beyond 1 member width away from the point of load application or geometric discontinuity, the average stress and strain assumptions hold (within 3%). Only if the member is in static equilibrium.

Stress Concentrations

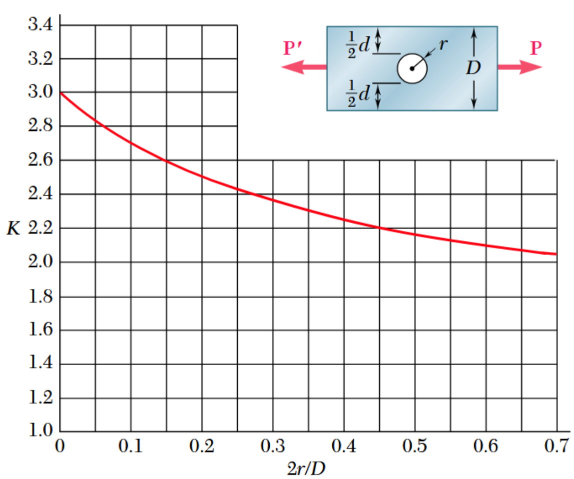
We can quantify the effect of stress concentrators using a *stress concentration factor* K. The stress concentration factor is strictly a function of geometry and load configuration where σ_{ave} is the local average normal stress at the critical area.

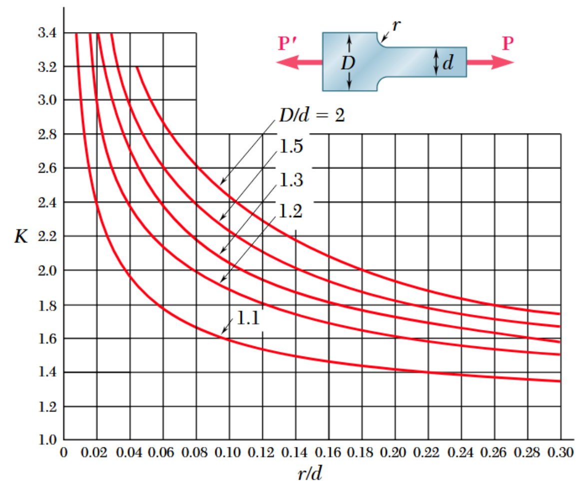
$$\boxed{K = \frac{\sigma_{\max}}{\sigma_{\text{ave}}}}$$

For example, near holes and fillets we have the following stress distributions



We determine the stress concentration factor K from tables and then use it to find the maximum stress at the critical area.





Note that the values of K are valid only for the elastic region where $\sigma_{\text{max}} < \sigma_{\text{yield}}$, and they are not valid for plastic deformation where the stress distribution is more uniform.

Plastic Deformation

When $\sigma_{\text{max}} > \sigma_{\text{yield}}$, we use the **bilinear elastoplastic material model**. This means that once the material reaches the yield stress, it begins to deform plastically. In the ideal elastoplastic model, the stress remains *constant* at the yield value while the strain continues to increase until failure.

If the load is removed while the material is in the plastic region, the unloading path follows a linear slope equal to the elastic modulus. The stress returns to zero along this line, but a permanent plastic strain remains.

- $\sigma < \sigma_{\text{yield}}$ (elastic region):

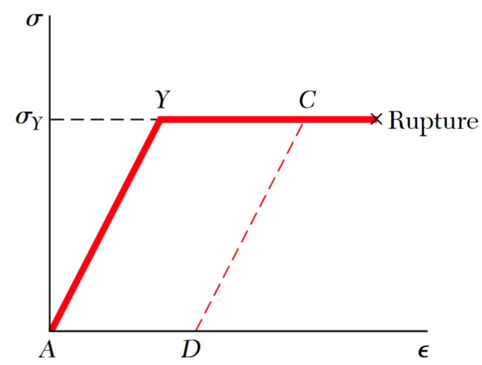
$$\sigma = E\varepsilon$$

- $\sigma \geq \sigma_{\text{yield}}$ (plastic region):

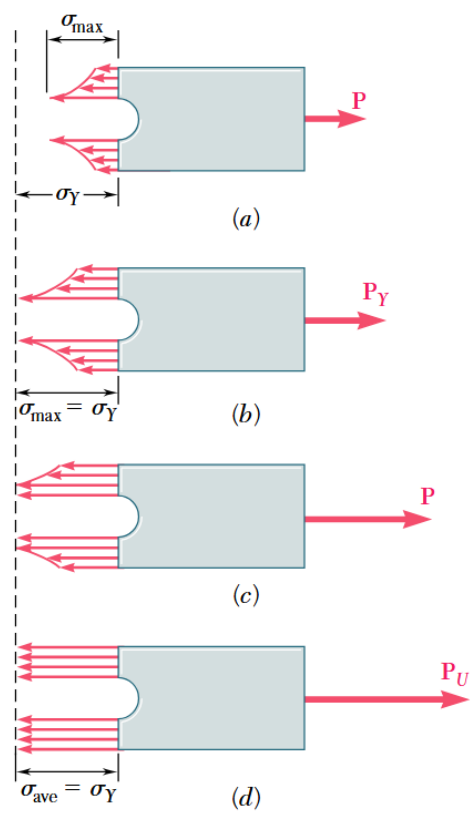
$$\sigma = \sigma_{\text{yield}} \quad (\text{increasing } \varepsilon)$$

- Unloading from plastic region:

$$\sigma = \sigma_{\text{yield}} - E(\varepsilon_{\text{max}} - \varepsilon) \quad (\text{decreasing } \varepsilon)$$



The stress distribution doesn't actually change in the plastic region, but it is offset more and more as the load increases (to maintain static equilibrium, the area under the curve must equal the applied load).



Under the bilinear elastoplastic material model, the stress cannot exceed the yield stress. So, as more of the cross-section yields, the stress distribution becomes increasingly uniform, approaching a fully plastic state. The maximum load therefore corresponds to the case in which the entire cross-section has reached the yield stress.

$$P_{\text{max}} = \sigma_{\text{yield}} A_{\text{min}}$$

Multiaxis *axial* Loading

In the generalized case where there are multiple axial loads in the x, y, and z directions, we can define the strain in each direction as:

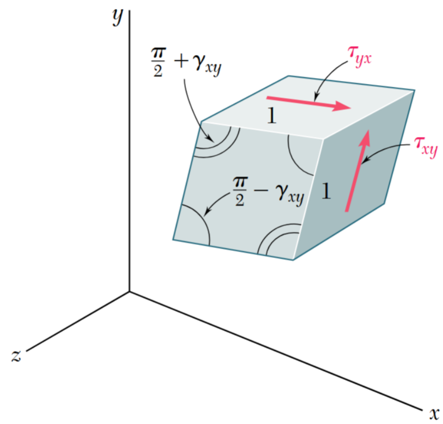
$$\varepsilon_x = \frac{\sigma_x}{E} - \left(\frac{\nu\sigma_y}{E} + \frac{\nu\sigma_z}{E}\right)$$
$$\varepsilon_y = \frac{\sigma_y}{E} - \left(\frac{\nu\sigma_x}{E} + \frac{\nu\sigma_z}{E}\right)$$
$$\varepsilon_z = \frac{\sigma_z}{E} - \left(\frac{\nu\sigma_x}{E} + \frac{\nu\sigma_y}{E}\right)$$

The strain in each direction the sum of the direct strain from the normal stress in that direction and Poisson's effect in the other two transverse directions.

Shear Stress and Strain

If a loading isn't purely axial, then we also have shear stress parallel to the surface. These shear stresses cause a deformation, called shear strain γ, which is an angular deformation in radians that describes how a cube of material shears into a parallelogram. The shear strain is related to the shear stress by the **shear modulus** or **modulus of rigidity** G.

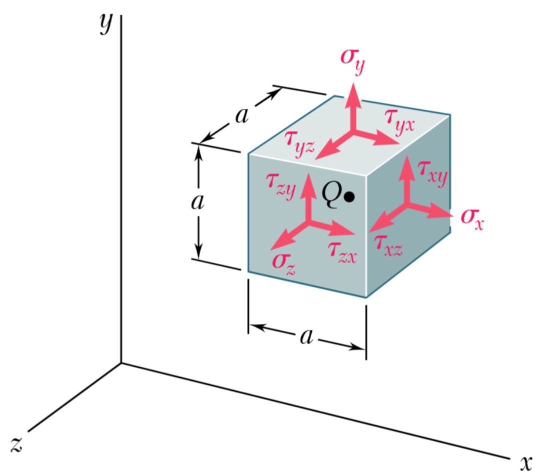
$$\boxed{\tau_{ij} = G\gamma_{ij}}$$



General Stress Strain Relations

Putting together the normal and shear stress-strain relations, we can express the general stress-strain relations in matrix form as normal stress strain describes the material changing size while shear stress strain describes the material changing shape (tilting of the cube).

$$\varepsilon_x = +\frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E} - \frac{\nu\sigma_z}{E}$$
$$\varepsilon_y = -\frac{\nu\sigma_x}{E} + \frac{\sigma_y}{E} - \frac{\nu\sigma_z}{E}$$
$$\varepsilon_z = -\frac{\nu\sigma_x}{E} - \frac{\nu\sigma_y}{E} + \frac{\sigma_z}{E}$$
$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$
$$\gamma_{xz} = \frac{\tau_{xz}}{G}$$
$$\gamma_{yz} = \frac{\tau_{yz}}{G}$$



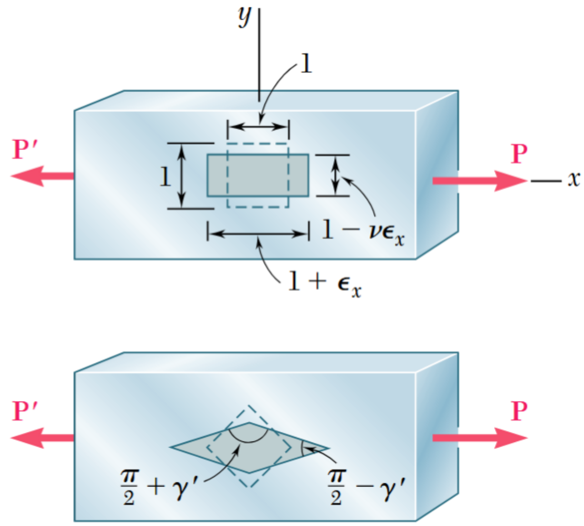
$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{xz} \\ \gamma_{yz} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{E}{G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{E}{G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{E}{G} \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{xz} \\ \tau_{yz} \end{bmatrix}$$

Note: These relationships are only valid for homogeneous, isotropic, linear elastic materials. Which means that the material properties are the same in all directions and the stress-strain relationship is linear (Hooke's law holds).

Normal and Shear Strains

Since the normal and shear stress change with the orientation of the cross-section, we can relate E, G, and ν using the following relation:

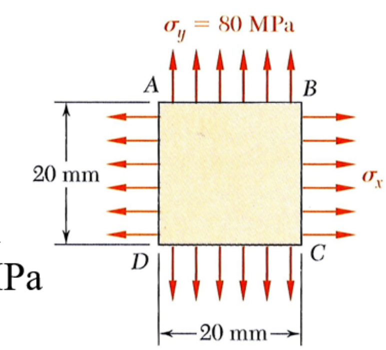
$$\boxed{\nu = \frac{E}{2G} - 1}$$

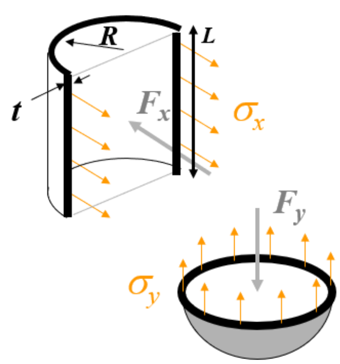


Pressure Vessels

Example P 2.67

- State of Stress of pressure vessel wall given σ_y=80MPa, σ_x=160MPa
- Goal: Find % length change of diagonal line across 20mm square
- Discussion on pressure vessels
- Approach
  - Direct computation of ε_x and ε_y
  - Geometry calculation





The pressure P inside the vessel will push outwards on the sides and ends, creating the forces F_x and F_y. We can relate these forces to the **hoop stress** σ_x and **axial stress** σ_y using force balances:

$$\sum F_x = 0 = -P(L2R) + \sigma_x(2tL) \quad \implies \sigma_x = \frac{PR}{t}$$

$$\sum F_y = 0 = -P(\pi R^2) + \sigma_y(\pi Rt) \quad \implies \sigma_y = \frac{PR}{2t}$$

$$\implies \sigma_x = 2\sigma_y$$

These normal stresses are redrawn onto the rectangle of material (if we unrolled the side of the vessel). If given Poisson's ratio and the modulus of elasticity, we can find the strains in the x and y directions using the general stress-strain relations.

$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E}$$

$$\varepsilon_y = -\frac{\nu\sigma_x}{E} + \frac{\sigma_y}{E}$$

Thus, the change in the diagonal will just be geometry:

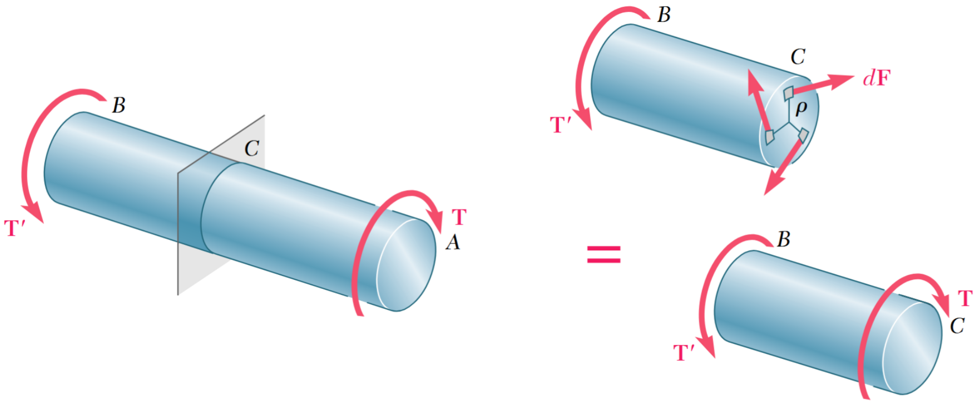
$$L_{x'} = L + L\varepsilon_x \quad \text{(changed length)}$$

$$L_{y'} = L + L\varepsilon_y \quad \text{(changed length)}$$

- **Original Diagonal (L_0):** Calculated using the Pythagorean theorem: $\sqrt{20^2 + 20^2}$.
- **Deformed Diagonal (L_f):** Calculated using the new side lengths: $\sqrt{(L_{x'})^2 + (L_{y'})^2}$.
- **% Length Change:** Calculated as $\frac{L_f - L_0}{L_0} \times 100$.

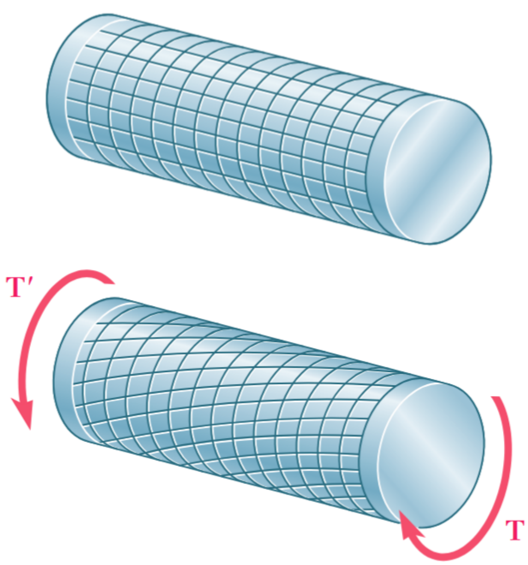
Torsion

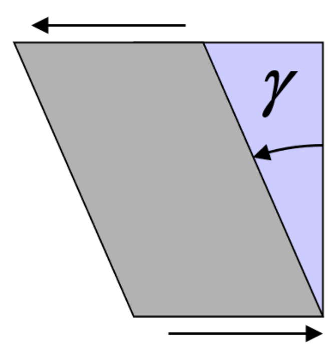
Torque is the result of shear stress. Consider the cross-section of a shaft under torsion, where T' is the internal torque at the cross-section:



Thus, torque can only be transmitted from shear stress in the tangential direction.

Deformation of Axisymmetric Shaft in Torsion
Similar to the intuition of shear strain from before, the cross-section doesn't change size but rather changes shape. Under torsion, the cross-section of the shaft remain the same and the diameter doesn't change but the cross-section twists, and the rectangles become parallelograms.

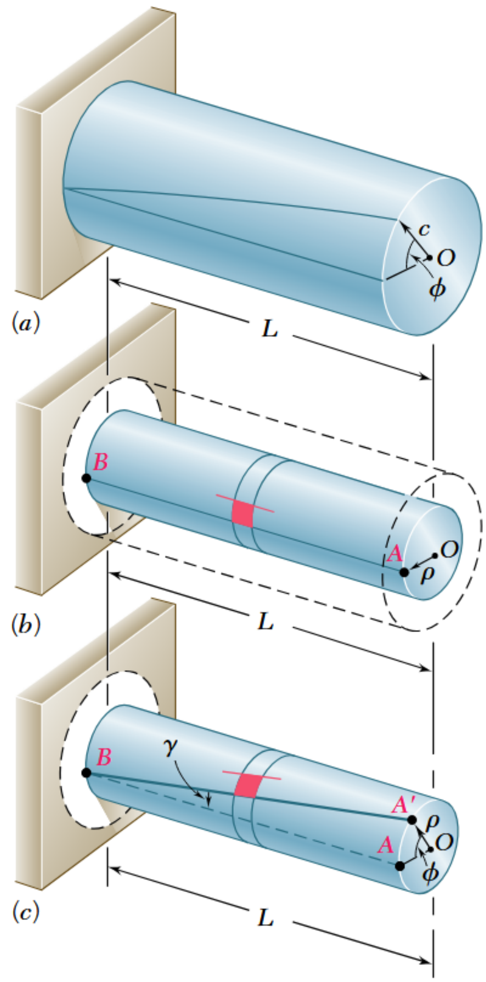




Twist Angle and Shear Strain

The relationship between the twist angle ϕ and the shear strain γ can be generalized for any radial position ρ in the cross-section (where c is the radius of the cross-section):

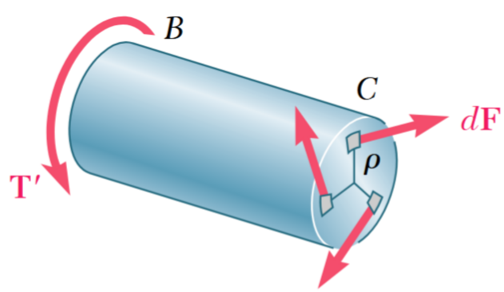
$$\boxed{\gamma = \frac{\rho\phi}{L}}$$



Torque and Twist Angle

Assuming a linear elastic material: $\tau = G\gamma$, we can relate the torque to the twist angle by integrating over the cross-section:

$$\begin{aligned} T &= \int \rho\, dF_\theta = \int \rho\tau\, dA = \int \rho G\gamma\, dA \\ &= \int \rho G \frac{\rho\phi}{L}\, dA = \frac{G\phi}{L} \underbrace{\int \rho^2\, dA}_{\substack{\text{Polar moment} \\ \text{of inertia } (J)}} = \frac{GJ\phi}{L}. \end{aligned}$$



$$\boxed{T = \frac{GJ\phi}{L}}$$

Polar Moment of Inertia
The polar moment of inertia J has units of length^4 (eg. m^4) and is based on the geometry of the cross-section. For a circular cross-section:

$$J = \int \rho^2 \, dA = \int_0^c \rho^2 2\pi\rho d\rho = \frac{\pi}{2}c^4$$

$$\boxed{J = \frac{\pi}{2}c^4 \quad \text{(round shaft)}}$$

Using superposition, we can find the polar moment of inertia for a hollow circular cross-section by subtracting the inner area from the outer area:

$$\boxed{J = \frac{\pi}{2}(c_o^4 - c_i^4) \quad \text{(hollow round shaft)}}$$

Shear Stress Distribution

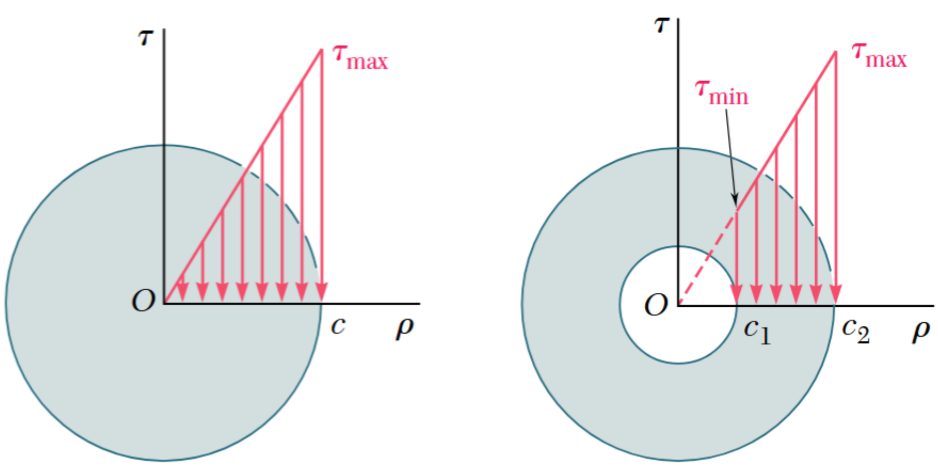
Rewriting the relationship between shear stress and shear strain, we find that τ is linearly proportional to the radial position ρ. Thus, it is max at the outer radius and zero at the center of the shaft.

$$\boxed{\tau = G\gamma = \frac{G\rho\phi}{L}}$$

$$\boxed{\tau_{\max} = \frac{Gc\phi}{L}}$$

Relating Torque to Shear Stress
We can rewrite the shear stress formula in terms of torque by:

$$\boxed{\tau = \frac{T\rho}{J} \implies \tau_{\max} = \frac{Tc}{J}}$$



Shaft Stiffness

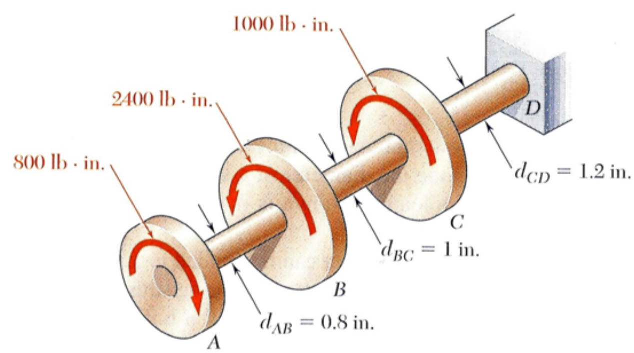
Like axial loading, we can define the stiffness of the shaft as the ratio of the load (torque) to the deformation (twist angle):

$$\boxed{k_T = \frac{T}{\phi} = \frac{GJ}{L}}$$

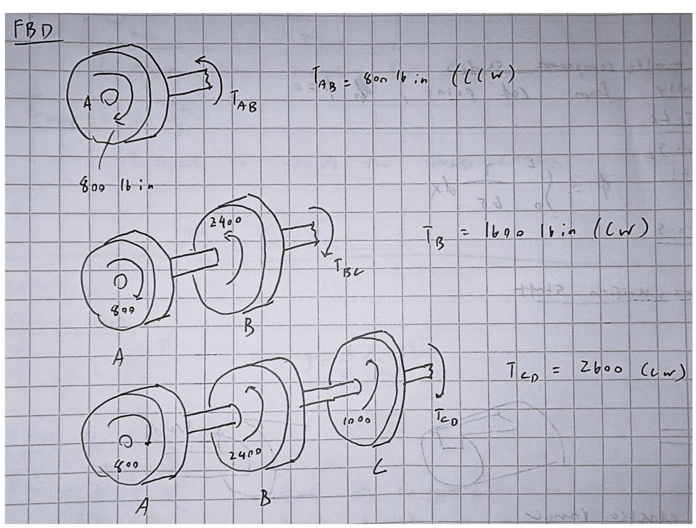
Multiple Torsional Loads

Ex P 3.10

- Hollow shafts (0.4" ID) with torques applied
- Find shaft in which max shear stress occurs
- Find maximum shear stress



Similar to axial deflection before, the torque T is the internal reaction torque at the cross-section, not the applied torque. Thus, we need to first solve for the internal torques using static equilibrium (FBD) and then apply the torsion formulas to find the shear stress and twist angle. (Keep track of direction of the twist angle!)



Total Shaft Twist
We can just sum up the total shaft twist using superposition:

$$\phi = \sum \frac{T_i L_i}{G_i J_i}$$

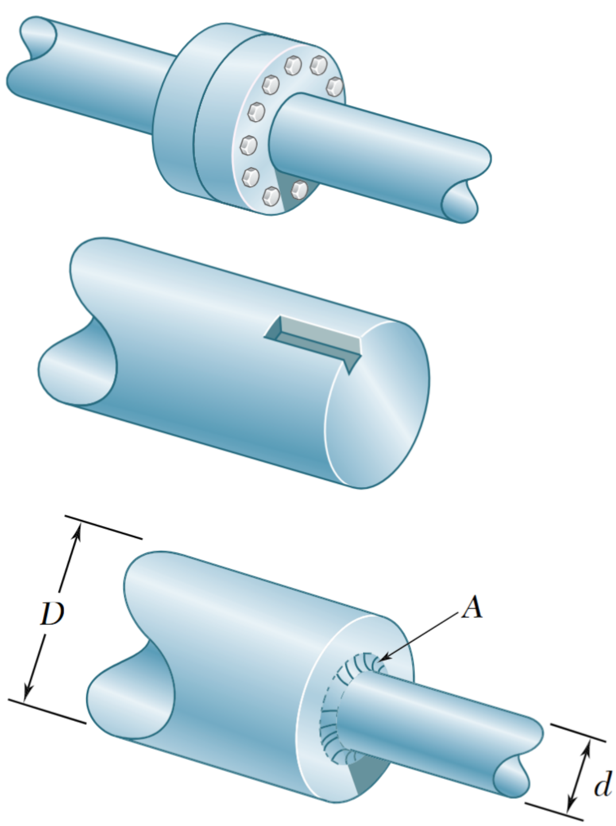
For a continuous shaft, we can integrate over the length of the shaft:

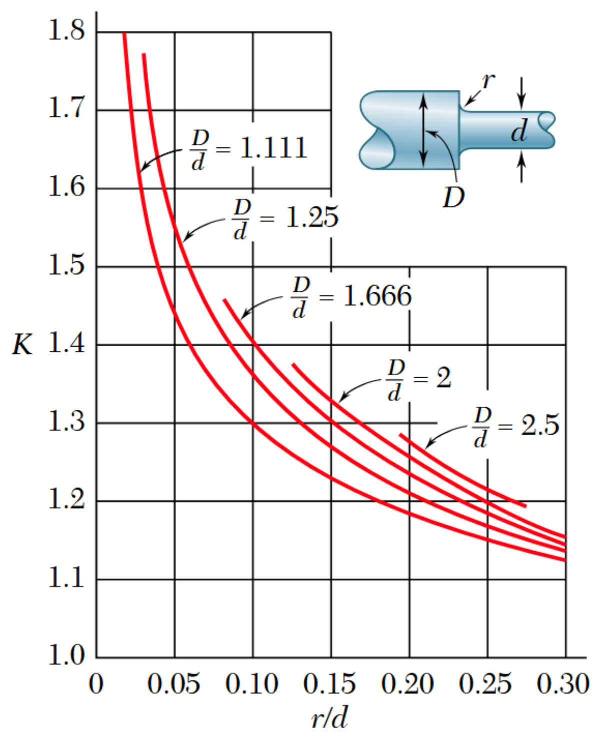
$$\phi = \int_0^L \frac{T}{GJ} dx$$

Stress Concentrations in Circular Shafts

Similar to axial loading, when we have a change in geometry (eg. hole, notch, fillet) or a point load, we have a stress concentration. We use the same stress concentration factor K to relate the max shear stress to the average shear stress at the critical area. Usually, we look at changes in shaft diameter, so the average shear stress is just the shear stress for a uniform shaft with the smaller diameter.

$$\boxed{\begin{aligned} \tau_{\text{max}} &= K\tau_{\text{max,uniform shaft}} \\ &= K\frac{Td}{2J}, \quad c = \frac{d}{2} \end{aligned}}$$





Shaft Power

Recall that the power transmitted by a rotating shaft is the product of the torque by its rotation rate.

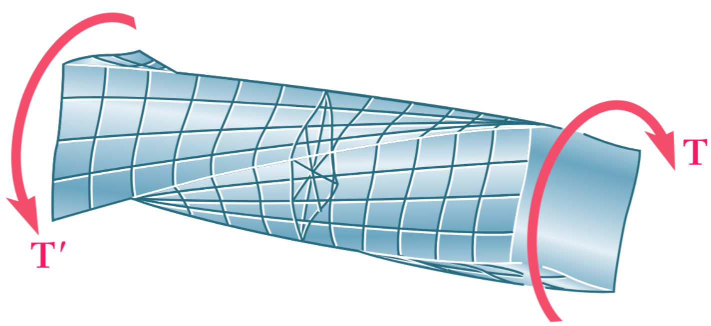
$$\boxed{\dot{W} = T\omega}$$

We will be assuming steady state where the angular velocity ω is constant, so we can still apply principles of static equilibrium to solve for the internal torques and then find the power transmitted by the shaft.

Now we are able to relate rotation to shear stress using power. Note that for a *fixed power*, a smaller angular velocity will require a larger torque, which will lead to a larger shear stress.

Torsion in Non-Circular Members

Lack of symmetry in the cross-section complicates torsion. The cross-sections of non-circular members don't just twist, but also warps. The corners have no shear stress since there's nothing to pull onto it, so it's a less efficient shape than a circular cross-section.



Torsion in Rectangular Bars

The max shear stress occurs on the centerline of the *wide faces* of the bar.

$$\boxed{\tau_{\max} = \frac{T}{c_1 a b^2}}$$

$$\boxed{\phi = \frac{TL}{c_2 a b^3 G}}$$

Where c_1 and c_2, like the stress concentration factor, are functions of the aspect ratio a/b and can be found in tables. Notice that the thickness b does more to resist torque than the width a.

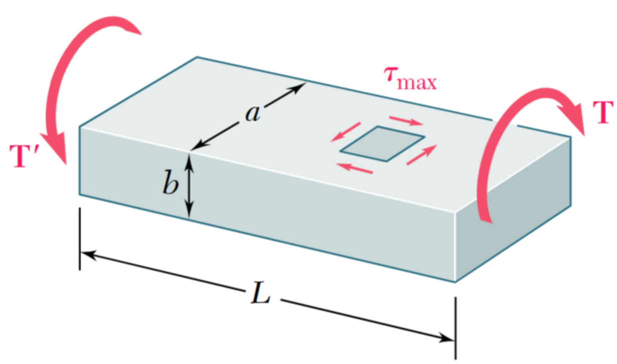
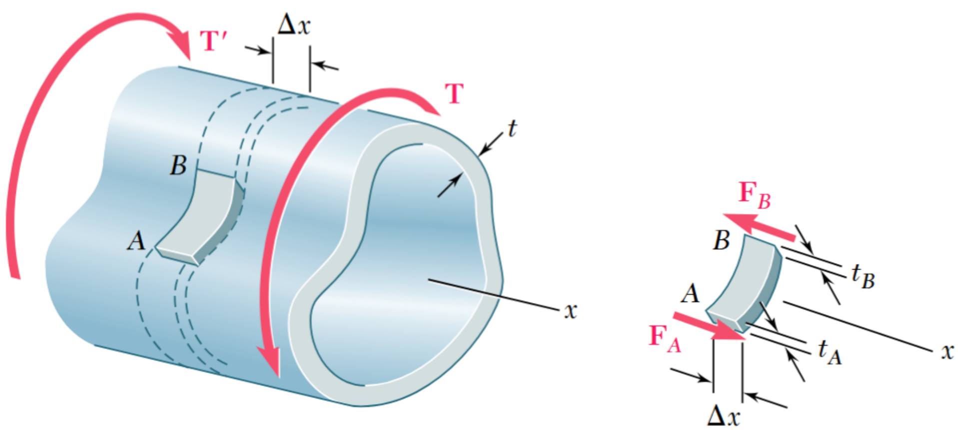


TABLE 3.1. Coefficients for Rectangular Bars in Torsion

a/b	c_1	c_2
1.0	0.208	0.1406
1.2	0.219	0.1661
1.5	0.231	0.1958
2.0	0.246	0.229
2.5	0.258	0.249
3.0	0.267	0.263
4.0	0.282	0.281
5.0	0.291	0.291
10.0	0.312	0.312
∞	0.333	0.333

Thin-Walled Hollow Shafts

Consider the following hollow shaft. Looking along the strip of width $\Delta x <<$ (thickness), we can see that because the forces F_A and F_B are balanced (due to equilibrium), we can define the shear flow q.



$$F_A = F_B$$

$$\tau_A t_A \Delta x = \tau_B t_B \Delta x \quad \Longrightarrow \quad \tau_A t_A = \tau_B t_B = \text{constant}$$

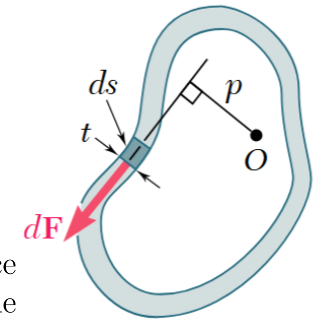
Thus, we define this constant as the shear flow q.

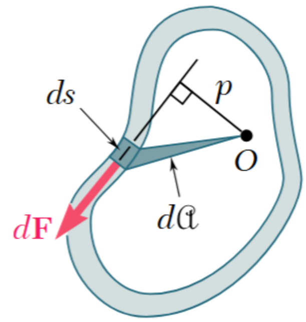
$$\boxed{q = \tau t}$$

Finding the Shear Stress in a Hollow Shaft

We will use the fact that the shear flow is constant across the cross-section to find the shear stress and twist angle. Like before, we can relate the torque to shear stress by integrating over the cross-section:

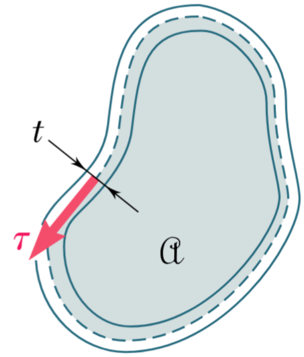
$$
\begin{aligned}
T &= \oint p\, dF \\
&= \oint p(\tau t\, ds) \\
&= q \oint 2\, da
\end{aligned}
$$





Notice we expressed the product $p\,ds$ as twice the area of the shaded triangle da, since the area of any triangle is just $\frac{1}{2} \times$ base $\times$ height, where the base is ds and the height is p. Thus, we can relate shear stress to torque by:

$$\boxed{\tau = \frac{T}{2ta}}$$



Where a is the area enclosed by the *shaft midline* of the cross-section (the line running in the center of the thickness).

Finding the Twist Angle in a Hollow Shaft

- Find ϕ, given $\tau = \dfrac{T}{2ta}$
- Approach:
  - Assume linear elastic
  - Work = strain energy (assuming statics)

Strain Energy Density:

$$u \equiv \int_0^\gamma \tau d\gamma = \int_0^\gamma G\gamma d\gamma = \frac{1}{2}G\gamma^2 = \frac{1}{2G}\tau^2$$

Linear Elastic $\implies T = k_T\phi$

Work of Torsion:

$$W = \int_0^\phi T d\phi = \int_0^\phi k_T\phi d\phi = \frac{1}{2}k_T\phi^2 = \boxed{\frac{1}{2}T\phi}$$

Strain Energy:

$$U = \int udV = \oint \frac{\tau^2}{2G}Ltds = \oint \frac{T^2}{2G4a^2t^2}Ltds = \boxed{\frac{T^2L}{2G4a^2}\oint \frac{ds}{t}}$$

Work = Elastic Energy in linear elastic shaft:

$$\implies \frac{1}{2}T\phi = \frac{T^2 L}{2G4a^2}\oint \frac{ds}{t}$$

$$\boxed{\phi = T\frac{L}{4Ga^2}\oint \frac{ds}{t}}$$

$$T = k_T\phi, \text{ where } k_T \equiv \frac{4Ga^2}{L\oint \dfrac{ds}{t}}$$

Here, ds refers to the path along the same midline of the shaft as before, and t is the thickness at that point along the path. Keep this in mind when evaluating the integral $\oint \frac{ds}{t}$.

<u>Note:</u> The criteria for thin-walled is that the thickness t is less than $\frac{1}{10}$ of the radius of curvature of the midline. For a circular shaft, this means $t < \frac{1}{10}c$.